Fiscal Year:	FY 2005	**Task Last Updated:**	FY 05/11/2006
PI Name:	Helble, Joseph J. Ph.D.		
Project Title:	High Capacity Thermally Regenerated Supported Amines for CO2 Capture		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	Joseph.J.Helble@Dartmouth.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	603-646-2238
Organization Name:	Darthmouth University		
PI Address 1:	Thayer School of Engineering		
PI Address 2:	8000 Cummings Hall		
PI Web Page:			
City:	Hanover	**State:**	NH
Zip Code:	03755-8000	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	08/02/2002	**End Date:**	09/01/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Zeolite sorbents are presently used for CO2 removal on long-term human-occupied space missions. These sorbents are regenerable, but the regeneration process is energy intensive. The focus of this project is therefore to investigate novel supported amine sorbents that can be regenerated at much lower temperatures and with lower energy than presently required by zeolites. These sorbents will be prepared by immobilizing amines on polymeric supports, chemically modifying the amines to maximize CO2 capture capacity, and demonstrating the efficacy of a temperature swing process as the regeneration step. The temperature swing process has the potential to provide a major reduction in the energy required for sorbent regeneration, an important benefit in space environments. Supported amines also have the potential to reduce the energy costs associated with the pumped liquid systems envisioned for terrestrial flue gas CO2 removal If successful, this project is ultimately expected to lead to the development of commercial systems for high capacity reduced cost CO2 removal, suitable for both enclosed habitats and terrestrial applications. Develop low cost		

	CO2 sorbents that require reduced energy consumption for thermal regeneration.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The focus of this project is to investigate novel supported amine sorbents that can be regenerated at much lower temperatures and with lower energy than presently required by zeolites. The proposed use of a temperature swing regeneration process has the potential to provide a major reduction in the energy required for sorbent regeneration, an important benefit in space environments. Supported amines also have the potential to reduce the energy costs associated with the pumped liquid systems envisioned for terrestrial flue gas CO2 removal If successful, this project is thus expected to lead to the development of commercial systems for high capacity reduced cost CO2 removal, suitable for both enclosed habitats and terrestrial applications.
Task Progress:	Supported amines have been shown to absorb CO2 cyclically under temperature swing absorption (TSA) conditions, and show a substantial decrease in desorption energy compared to zeolite materials. Supported amines may therefore be a viable alternative for cyclic capture of CO2 on long-term space missions where minimal energy expenditure is a critical consideration. An experimental system was designed and constructed for testing the supported amines for CO2 cyclic capacity under temperature swing conditions. In a temperature swing system, the bed is heated to a specific desorption temperature and pressure is reduced by exposure to vacuum conditions to desorb the CO2. The system used in this study consisted of two absorbing beds filled as a packed bed reactor with supported amine material. The beds absorbed and desorbed CO2 cyclically and sequentially. Usually 15-20 cycles were completed before the beds desorbed as much as they absorbed and reached a steady state of operation. Steady state efficiency was calculated for varying conditions of the system and statistically analyzed for the importance of variables considered. Three amines were tested for cyclic CO2 capacity under TSA as part of this project: triethylenetetramine (TETA), tetraethylenepentamine (TEPAN), and ethyleneamine (E-100). TEPAN and E-100 have been modified by reaction with acrylonitrile to form secondary amines from primary amines. These three liquid amines were chosen for their low vapor pressure and high secondary amine content. A low vapor pressure minimizes amine losses at high temperatures and low system pressures experienced in the TSA process. The high amine content is a requirement because amines reversibly react with CO2 and are responsible for the capture of CO2 in the TSA system. The high amine content allows more CO2 to be captured per mass of amine. TETA was found to have too high a vapor pressure at the required desorption conditions and amine losses were substantial. TEPAN and E-100, on the other hand, had such low vapor pressures that amine losses were undetected. A fractional factorial statistical experiment was conducted on the TEPAN and E-100 amines in order to determine the importance of seven variables to the system. These included cycle time (15-20 min), CO2 inlet concentration (6-9mmHg), absorption temperature (25-30°C), residence time of the gas through the bed (1.7-3sec), inlet dew point (0-25°C), desorption temperature (50-60?C) and desorption pressure (26-93mmHg). Important variables to the TEPAN molecule were found to be the inlet concentration of CO2 (12% importance), the absorption temperature (6%), desorption temperature (19%), and desorption pressure (61%). For the E-100 molecule, only desorption pressure was important to the system. A full factorial experiment expanded on the relationship between desorption temperature and pressure for the E-100 molecule in an extended range. Desorption temperature was varied between 40, 50, and 60°C and desorption pressure was expanded to include 26, 51, and 93mmHg. Over this range, desorption temperature was the most significant to the results, contributing 61% of the importance to the system while desorption pressure contributed 33%. Three new amines were also considered as possible alternatives to TEPAN and E-100. These included N,N'-bis(2-hydroxyethyl)-ethylenediamine (BED), an acrylonitrile modified 1-(2-aminoethyl)piperazine (AEP), and cyclam. The same experimental setup described above was used to examine total CO2 capacity and compare these to the original TEPAN supported amine. The results indicate that there was no change in total capacity with the addition of BED or cyclam. With AEP, there was a decrease of less than half capacity of the original material. No cyclic testing was conducted on any of the three amines. PMMA, TEPAN and modified E-100 were tested at temperatures of 100-200°C for thermal stability in a gas chromatograph / mass spectrometer (GC/MS). Decomposition of PMMA begins at 150°C and increases in extent at 200°C. Acrylonitrile begins to appear at 150°C. The loss of acrylonitrile may hinder the capture ability of CO2 if the beads are left at these temperatures for extended periods. At 200°C, the decomposition of the amines becomes excessive. Because the proposed operation temperature is 60°C, however, thermal decomposition of PMMA, TEPAN, or modified E-100 should not be a concern. In summary, supported amines utilized under temperature swing conditions have been demonstrated to be a viable alternative to the current zeolite sorbent system of CO2 capture. Supported amines offer comparatively low desorption temperatures and energy requirements, high capacity for CO2, and high tolerance of humidity. In addition, this study has shown that increasing the thermal energy can significantly reduce the vacuum pump requirements of the desorption step. Lastly, the amines have been shown to be thermally stable at the 60°C system condition.
Bibliography Type:	Description: (Last Updated: 08/11/2011)
Papers from Meeting Proceedings	Schladt M, Filburn T, Helble JJ. "Advanced studies of supported amine sorbents for CO2 removal." International Conference on Environmental Systems (ICES), Rome, Italy, July 2005. International Conference on Environmental Systems (ICES), Rome, Italy, July 2005. SAE Paper 2005-01-2943. http://dx.doi.org/10.4271/2005-01-2943 , Jul-2005